

UNIT-11

- * A 2D geometric transformation is an operation that changes the position, size, orientation, or shape of an object in a two-dimensional coordinate system.
- * Transformation is performed by multiplying the coordinates of an object with a transformation matrix.

Types of 2D Transformation

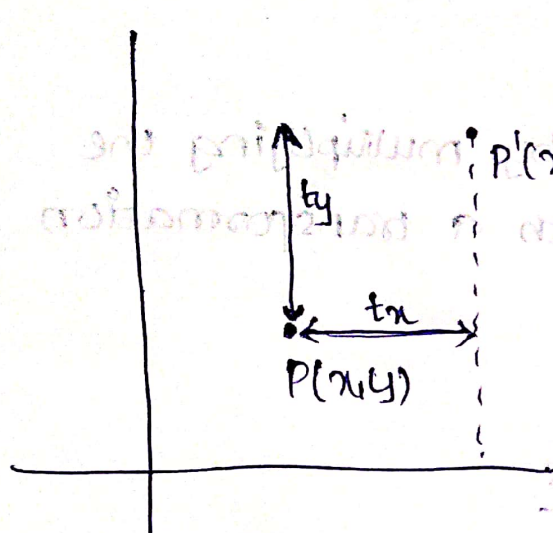
- (1) Translation
- (2) Scaling
- (3) Rotation
- (4) Reflection
- (5) Shearing

* Transformations are used to :

- (1) Move an object from one location to another.
- (2) Increase or decrease the size of an object.
- (3) Rotate an object to any angle.
- (4) Create mirror images.
- (5) Tilt or slant an object.
- (6) Perform animation and modeling.

(1) Translation

* It moves an object from one position to another without changing its size, shape or orientation.



$$\begin{aligned}x' &= x + tx \\ y' &= y + ty\end{aligned}$$

$$P' = P + T$$

~~Matrix Form:~~

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} x \\ y \end{bmatrix} + \begin{bmatrix} tx \\ ty \end{bmatrix}$$

Ex : Polygon $\Rightarrow A(2, 5), B(7, 10), C(10, 12)$.

$$tx = 2, ty = 2$$

$$\begin{aligned}A' \Rightarrow x' &= 2 + 2 = 4 \\ y' &= 5 + 2 = 7 \\ \Rightarrow A' &= (4, 7)\end{aligned}$$

$$\begin{aligned}B' \Rightarrow x' &= 7 + 2 = 9 \\ y' &= 10 + 2 = 12 \\ B' &= (9, 12)\end{aligned}$$

$$\begin{aligned}C' \Rightarrow x' &= 10 + 2 = 12 \\ y' &= 12 + 2 = 14 \\ C' &= (12, 14)\end{aligned}$$

* In translation, every point of the object is moved by the same distance in the same direction.

* The amount by which an object moves is represented by a translation vector $\Rightarrow T = (tx, ty)$

(2) Scaling

- * It changes the size of an object.
- * It can make an object larger (enlarge) or smaller (reduce) without changing its basic shape.
- * During scaling, the coordinates of every point are multiplied by scaling factors.
- * Unlike translation, which changes only the position, scaling changes the dimensions of the object.
- * ~~Sac~~ Scaling factor $\Rightarrow$
 - $S_x \rightarrow$ scaling factor along x-axis
 - $S_y \rightarrow$ scaling factor along y-axis.

* Types

1. Uniform scaling $\Rightarrow S_x = S_y$
 2. Non-uniform scaling $\Rightarrow S_x \neq S_y$
- * Cases :
1. Enlargement / Increase $\Rightarrow S_x > 1, S_y > 1$
 2. Reduction $\Rightarrow 0 < S_x < 1, 0 < S_y < 1$
 3. No change $\Rightarrow S_x = 1, S_y = 1$.

* Suppose the original point is $P(x, y)$

After scaling $\Rightarrow P'(x', y')$

$$\begin{aligned} x' &= S_x x \\ y' &= S_y y \end{aligned}$$

Example : $P = (3, 4)$, $S_x = 2$, $S_y = 3$

New coordinates $\Rightarrow x' = 2 \times 3 = 6$

$$y' = 3 \times 4 = 12$$

$$\therefore P' = (6, 12)$$

Ex 1) Suppose a triangle has vertices A , B , C
 $A(1, 2)$, $B(4, 2)$, $C(2, 5)$

Scaling Factors $S_x = 2$, $S_y = 2$

Calculate Each vertex,

$$A' \Rightarrow x' = 1 \times 2 = 2$$

$$y' = 2 \times 2 = 4$$

$$\therefore A' = (2, 4)$$

$$B' \Rightarrow x' = 4 \times 2 = 8$$

$$y' = 2 \times 2 = 4$$

$$B' = (8, 4)$$

$$C' \Rightarrow x' = 2 \times 2 = 4$$

$$y' = 5 \times 2 = 10$$

$$C' = (4, 10)$$

(3) Rotation

* It is a geometric transformation that turns an object through a specified angle about a fixed point.

* During rotation :

- The shape of the object remains unchanged.
- The size of the object remains unchanged.
- Only the orientation (direction) of the object changes.

*

$$\cos \alpha = \frac{x}{r} \Rightarrow x = r \cos \alpha$$

$$\sin \alpha = \frac{y}{r} \Rightarrow y = r \sin \alpha$$

$$\cos(\theta + \alpha) = \frac{x'}{r} \Rightarrow x' = r \cos(\theta + \alpha)$$

$$x' = r (\cos \theta \cos \alpha - \sin \theta \sin \alpha)$$

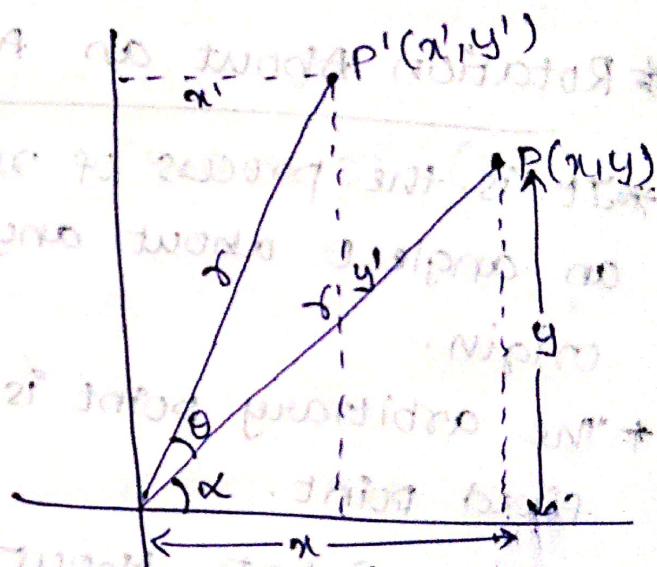
$$x' = r \cos \theta \cos \alpha - r \sin \theta \sin \alpha$$

$$\boxed{x' = x \cos \theta - y \sin \theta}$$

$$\sin(\theta + \alpha) = \frac{y'}{r} \Rightarrow y' = r \sin(\theta + \alpha)$$

$$y' = r \cos \theta \sin \alpha + r \sin \theta \cos \alpha$$

$$\boxed{y' = y \cos \theta + x \sin \theta}$$



Example $\Rightarrow P(4, 3)$ angle $(\theta) = 45^\circ$

$$x' = x \cos \theta - y \sin \theta$$
$$= x \cos(45^\circ) - y \sin(45^\circ)$$

$$= \frac{4}{\sqrt{2}} - \frac{3}{\sqrt{2}} = \frac{1}{\sqrt{2}}$$

$$y' = y \cos \theta + x \sin \theta$$

$$= \frac{3}{\sqrt{2}} + \frac{4}{\sqrt{2}} = \frac{7}{\sqrt{2}}$$

$$\therefore P' = \left(\frac{1}{\sqrt{2}}, \frac{7}{\sqrt{2}} \right)$$

(4) Shearing

- * It is a geometric transformation that slants the shape of an object along the x-axis or y-axis without changing its area significantly.
- * In shearing, one coordinate changes in proportion to the other coordinate.
- * Unlike scaling, which changes the size, or rotation, which changes the orientation, shearing distorts the shape of the object by shifting its sides.
- * x-shear (Horizontal shearing)

- In x-shear, only the x-coordinate changes, while the y-coordinate remains unchanged.

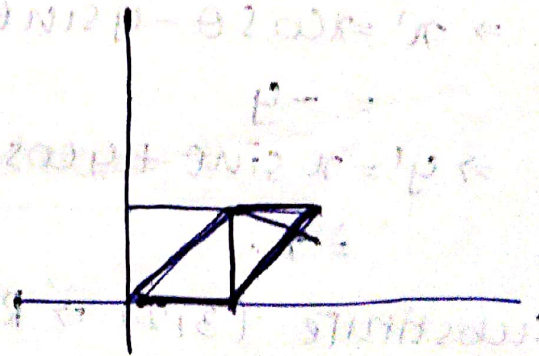
$$x' = x + sh_x \times y$$
$$y' = y$$

Ex: $P = (4, 3)$ $sh_x = 2$

$$x' = 4 + (2 \times 3) = 10$$

$$y' = 3$$

$$\therefore P' = (10, 3)$$



* Y-shear (Vertical shearing)

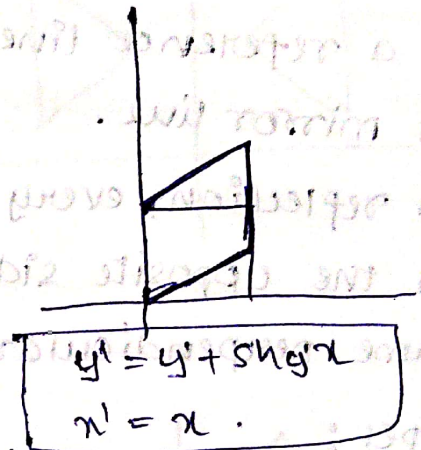
In Y-shear, only the y-coordinate changes, while the x-coordinate remains unchanged.

Ex: $P = (4, 3)$, $Sh_y = 1.5$

$$x' = 4$$

$$y' = 3 + (1.5 \times 4) = 9$$

$$P' = (4, 9)$$



* $Sh = 0 \Rightarrow$ No shearing occurs.

$Sh > 0 \Rightarrow$ The object slants towards the +ve axis.

$Sh < 0 \Rightarrow$ The object slants towards the -ve axis.

(5) Reflection

* It is a geometric transformation that produces the mirror image of an object with respect to a reference line called axis of reflection or mirror line.

* In reflection, every point of the object is placed on the opposite side of the mirror line at the same perpendicular distance from it.

* Types :

(i) x-axis

(ii) y-axis

(iii) origin

(iv) line $y = x$

(v) line $y = -x$

(i) Reflection About x-axis

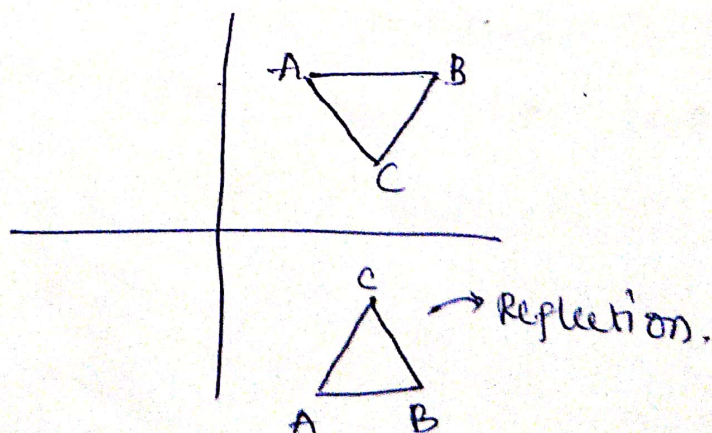
The object is reflected about x-axis

• The x-coordinate remains unchanged

• The y-coordinate changes its sign.

$$x' = x$$

$$y' = -y$$

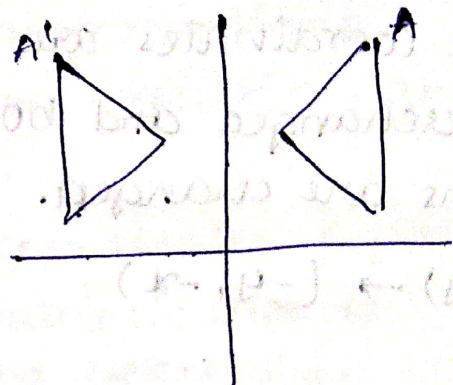


(ii) Reflection About y-axis.

The object is reflected about y-axis.

- The y-coordinates remain same
- The x-coordinates change its sign.

$$(x, y) \rightarrow (-x, y).$$

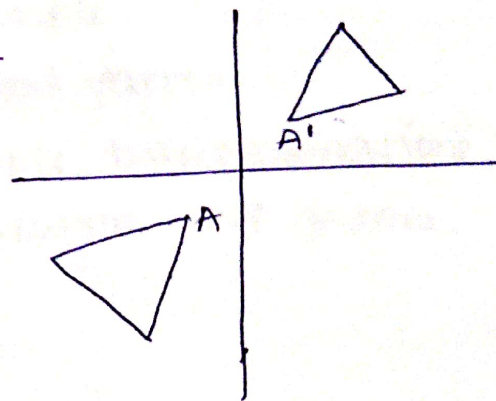


(iii) Reflection about origin.

Reflection takes place about the origin (0,0)

Both coordinates change their signs.

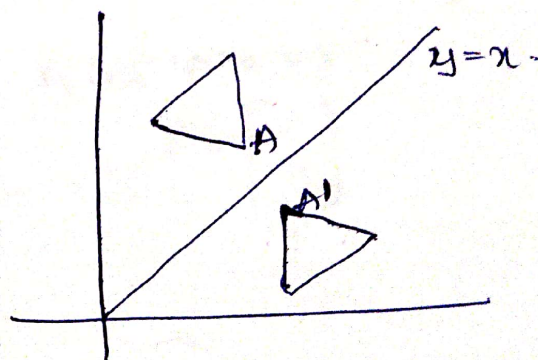
$$(x, y) \rightarrow (-x, -y).$$



(iv) Reflection about the line $y=x$.

- Reflection takes place about the line $y=x$
- The x and y coordinates are interchanged.

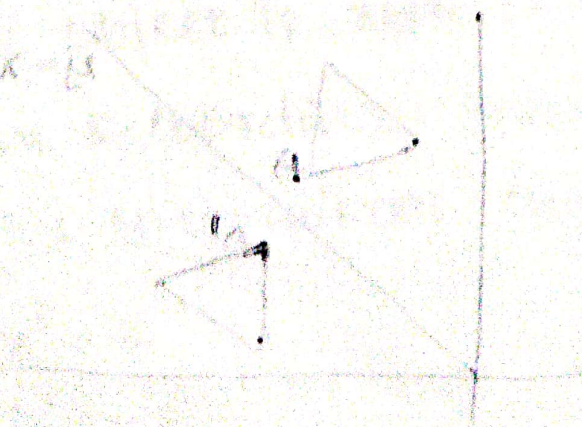
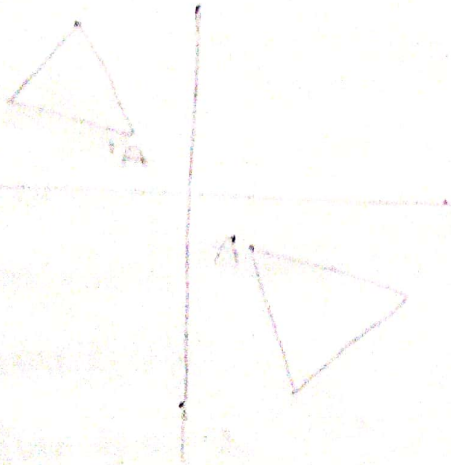
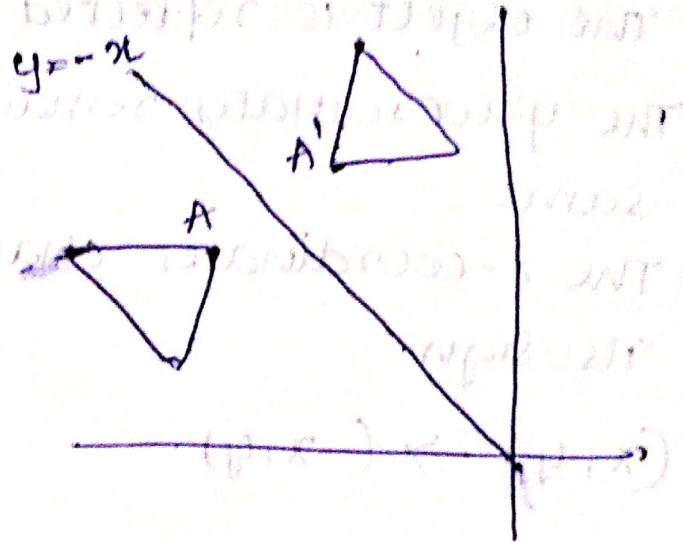
$$(x, y) \rightarrow (y, x)$$



(v) Reflection about the line $y = -x$.

The coordinates are interchanged and both signs are changed.

$$(x, y) \rightarrow (-y, -x)$$



→ Matrix Representation

* It is a mathematical method used to represent and perform geometric transformations.

(translation, scaling, rotation, reflection, and shearing) using matrices.

* Instead of calculating the coordinates separately everytime, we use matrix multiplication to transform objects quickly and efficiently.

* Why do we need use matrix representation?

- makes calculations simple.
- Performs transformations faster.
- Easy to combine multiple transformations
- used in all graphics software and games.

Types of Matrices

(a) Point Matrix

A point is represented as a column vector,

For a 2D point :

$$P \begin{bmatrix} x \\ y \end{bmatrix}$$

Point A(3,4)

$$P = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$$

$$\begin{bmatrix} x \\ y \end{bmatrix} \begin{bmatrix} x & y \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} x \\ y \end{bmatrix}$$

(2) Transformation Matrix

A transformation is represented by a square matrix.

(i) Translation

$$x' = x + t_x$$

$$y' = y + t_y$$

$$\Rightarrow P' = P + T$$

$$\rightarrow \begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} x \\ y \end{bmatrix} + \begin{bmatrix} t_x \\ t_y \end{bmatrix}$$

(ii) Scaling Matrix

$$x' = S_x x$$

$$y' = S_y y$$

$$\Rightarrow P' = S \times P$$

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} S_x & 0 \\ 0 & S_y \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

(iii) Rotation Matrix

$$x' = x \cos \theta - y \sin \theta$$

$$y' = x \sin \theta + y \cos \theta$$

$$R = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$$

$$\Rightarrow P' = R \times P$$

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

(iv) Shearing Matrix

(a) X-shear

$$x' = x + sh_x y$$

$$y' = y$$

$$P' = sh_x \times P$$

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} 1 & sh_x \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

(b) y -shearing.

$$x' = x$$

$$y' = y + sh_y x$$

$$P' = sh_y \times P$$
$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ sh_y & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

(*) Reflection Matrix

(a) Reflection about x -axis

$$x' = x$$

$$y' = -y$$

$$P' = R_x \times P$$

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

(b) Reflection about y -axis

$$x' = -x$$

$$y' = y$$

$$P' = R_y \times P$$

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

(c) Reflection about the origin.

$$x' = -x$$

$$y' = -y$$

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

(d) Reflection about the line $y = x$

$$x' = y$$

$$y' = x$$

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

(e) Reflection about the line $y = -x$

$$x' = -y$$

$$y' = -x$$

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} 0 & -1 \\ -1 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

→ Homogeneous Coordinates

* Homogeneous coordinates are an extension of the normal Cartesian coordinate system in which one extra coordinate (w) is added to every point.

In 2D, a point (x, y) is represented as

$$(x, y, 1) \quad \text{or} \quad P = \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

Extra coordinate 1 is called the homogeneous coordinate.

→ Homogeneous Transformation Matrices

Translation. General transformation equation is $P' = TP$



$\begin{bmatrix} x \\ y \\ 1 \end{bmatrix} \rightarrow P \rightarrow \text{original point}$

$T \rightarrow 3 \times 3$ transformation matrix

$P' \rightarrow \text{Transformed point}$

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

(1) Translation

$$P' = TP$$

$$P' = \begin{bmatrix} 1 & 0 & T_x \\ 0 & 1 & T_y \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

$$\begin{cases} x' = x + T_x \\ y' = y + T_y \end{cases}$$

$$\therefore P' = \begin{bmatrix} x + T_x \\ y + T_y \\ 1 \end{bmatrix}$$

(2) Scaling

$$P' = SP$$

$$= \begin{bmatrix} S_x & 0 & 0 \\ 0 & S_y & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

$$\begin{cases} x' = S_x x \\ y' = S_y y \end{cases}$$

$$\therefore P' = \begin{bmatrix} S_x x \\ S_y y \\ 1 \end{bmatrix}$$

(3) Rotation

$$P' = RP$$

$$= \begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = \begin{bmatrix} x \cos\theta - y \sin\theta \\ x \sin\theta + y \cos\theta \\ 1 \end{bmatrix}$$

$$\begin{cases} x' = x \cos\theta - y \sin\theta \\ y' = x \sin\theta + y \cos\theta \end{cases}$$

(4) Reflection about x-axis.

$$P' = RP$$

$$= \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = \begin{bmatrix} x \\ -y \\ 1 \end{bmatrix}$$

$$\begin{aligned} x' &= x \\ y' &= -y \end{aligned}$$

Reflection about y-axis

$$P' = RP$$

$$= \begin{bmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = \begin{bmatrix} -x \\ y \\ 1 \end{bmatrix}$$

$$\begin{aligned} x' &= -x \\ y' &= y \end{aligned}$$

Reflection about Origin.

$$P' = RP$$

$$= \begin{bmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = \begin{bmatrix} -x \\ -y \\ 1 \end{bmatrix}$$

$$\begin{aligned} x' &= -x \\ y' &= -y \end{aligned}$$

Reflection about line $y=x$

$$P' = RP$$

$$= \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = \begin{bmatrix} y \\ x \\ 1 \end{bmatrix}$$

$$\begin{aligned} x' &= y \\ y' &= x \end{aligned}$$

Reflection about line $y = -x$

$$P' = RP$$

$$x' = -y$$

$$y' = -x$$

$$= \begin{bmatrix} 0 & -1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = \begin{bmatrix} -y \\ -x \\ 1 \end{bmatrix}$$

(5) X-shearing

$$P' = Sh_x P$$

$$x' = x + sh_x y$$

$$y' = y$$

$$= \begin{bmatrix} 1 & sh_x & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = \begin{bmatrix} x + sh_x y \\ y \\ 1 \end{bmatrix}$$

(6) Y-shearing

$$P' = Sh_y P$$

$$x' = x$$

$$y' = y + sh_y x$$

$$= \begin{bmatrix} 1 & 0 & 0 \\ sh_y & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = \begin{bmatrix} x \\ y + sh_y x \\ 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 0 \\ sh_y & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ sh_y & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

→ Composite Transformation.

A composite transformation is the process of applying two or more geometric transformations one after another to an object.

Instead of applying each transformation separately, we combine all transformation matrices into one composite transformation matrix.

⇒ Translations

If 2 successive translation vectors (t_{x1}, t_{y1}) and (t_{x2}, t_{y2}) are applied to a coordinate position P , the final transformed location P' is calculated as

$$P' = T(t_{x2}, t_{y2}) \cdot \{ T(t_{x1}, t_{y1}) \cdot P \}$$

$$\boxed{P' = \{ T(t_{x2}, t_{y2}) \cdot T(t_{x1}, t_{y1}) \} \cdot P}$$

$$= \begin{bmatrix} 1 & 0 & t_{x2} \\ 0 & 1 & t_{y2} \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 & 0 & t_{x1} \\ 0 & 1 & t_{y1} \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & t_{x1} + t_{x2} \\ 0 & 1 & t_{y1} + t_{y2} \\ 0 & 0 & 1 \end{bmatrix}$$

$$P' = \begin{bmatrix} 1 & 0 & t_{x1} + t_{x2} \\ 0 & 1 & t_{y1} + t_{y2} \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

⇒ Scalings

$$P' = \{S(sx_1, sy_1) \cdot S(sx_2, sy_2)\} \cdot P$$

$$\Rightarrow \begin{bmatrix} sx_2 & 0 & 0 \\ 0 & sy_2 & 0 \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} sx_1 & 0 & 0 \\ 0 & sy_1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} sx_1 \cdot sx_2 & 0 & 0 \\ 0 & sy_1 \cdot sy_2 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\Rightarrow P' = \begin{bmatrix} sx_1 \cdot sx_2 & 0 & 0 \\ 0 & sy_1 \cdot sy_2 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

⇒ Rotation.

$$P' = P \{ R(\theta_1) \cdot R(\theta_2) \}$$

$$R(\theta_1) \cdot R(\theta_2) = \begin{bmatrix} \cos\theta_1 & \sin\theta_1 & 0 \\ -\sin\theta_1 & \cos\theta_1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \cos\theta_2 & \sin\theta_2 & 0 \\ -\sin\theta_2 & \cos\theta_2 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} \cos\theta_1 \cos\theta_2 - \sin\theta_1 \sin\theta_2 & \cos\theta_1 \sin\theta_2 + \sin\theta_1 \cos\theta_2 & 0 \\ -\sin\theta_1 \cos\theta_2 - \sin\theta_2 \cos\theta_1 & -\sin\theta_1 \sin\theta_2 + \cos\theta_1 \cos\theta_2 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} \cos(\theta_1 + \theta_2) & \sin(\theta_1 + \theta_2) & 0 \\ -\sin(\theta_1 + \theta_2) & \cos(\theta_1 + \theta_2) & 0 \end{bmatrix} = R(\theta_1 + \theta_2).$$

⇒ TRANSFORMATIONS BETWEEN COORDINATE SYSTEM

Transformation between coordinate systems is the process of converting the coordinates of a point from one coordinate system to another.

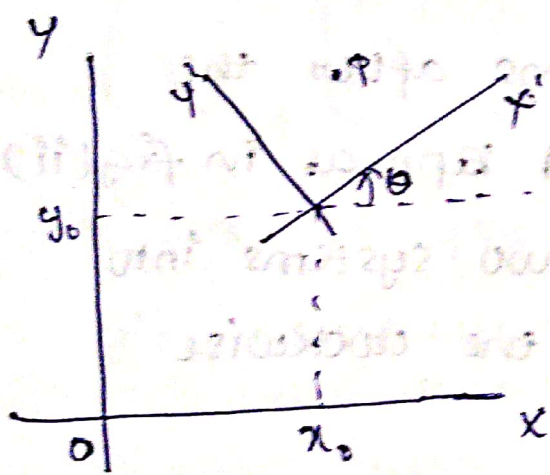


fig (i)

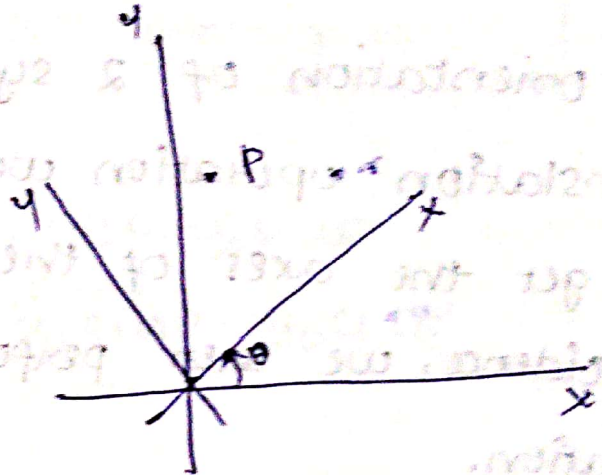


fig (ii)

~~The coordinate~~
The above figure (i) shows 2 cartesian systems, with coordinate origins at $(0,0)$ and (x_0, y_0) and with an orientation angle θ between x and x' axis.
To transform object descriptions from xy coordinates to $x'y'$ coordinates, we need to set up a transformation that superimposes the $x'y'$ axis onto the xy axis. This is done in 2 steps.

- (1) Translate so that the origin (x_0, y_0) of $x'y'$ system is moved to the origin of the xy system.

(2) Rotate the x' axis onto the x -axis.

→ Translation of the coordinate origin is expressed with the matrix operation.

$$T(-x_0, -y_0) = \begin{bmatrix} 1 & 0 & -x_0 \\ 0 & 1 & -y_0 \\ 0 & 0 & 1 \end{bmatrix}$$

the orientation of 2 systems after the translation operation would appear in fig(11) to get the axes of the two systems into coincidence, we then perform the clockwise rotation.

$$R(-\theta) = \begin{bmatrix} \cos\theta & \sin\theta & 0 \\ -\sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Concatinating these 2 transformation matrices gives us the complete composite matrix for transforming object descriptions from xy system to $x'y'$ system :

$$M = R(-\theta)T(-x_0, -y_0)$$

$$M = \begin{bmatrix} \cos\theta & \sin\theta & 0 \\ -\sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & -x_0 \\ 0 & 1 & -y_0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} \cos\theta & \sin\theta & -x_0\cos\theta - y_0\sin\theta \\ -\sin\theta & \cos\theta & x_0\sin\theta - y_0\cos\theta \\ 0 & 0 & 1 \end{bmatrix}$$

$$x' = x \cos\theta + y \sin\theta - x_0 \cos\theta - y_0 \sin\theta$$

$$y' = -x \sin\theta + y \cos\theta + x_0 \sin\theta - y_0 \cos\theta$$